

- 2 450 students were asked how they travelled to school on Monday.
 Each student walked or travelled by bus or travelled by car or travelled by bicycle.
 Each student used just one method of travel.

One of these students is chosen at random.

The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

$$0.20 + x + 2x + 0.26 = 1$$

$$3x + 0.46 = 1$$

$$3x = 1 - 0.46$$

$$\frac{3x}{3} = \frac{0.54}{3}$$

$$x = 0.18$$

$$\begin{aligned} \therefore \text{Car} &= 2x \\ &= 2 \times 0.18 = 0.36 \end{aligned}$$

$$\begin{aligned} \therefore \text{No of students travelling} &= 0.36 \times 450 \\ \text{by car} &= \underline{\underline{162}} \end{aligned} \quad 162 \text{ students}$$

(Total for Question 2 is 4 marks)